Accelerated Math II/III Name_____

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HW #15 - Inverse Relations & Functions

Date_____ Period____

State if the given functions are inverses. (Use composition of functions.)

1) $g(x) = \frac{3}{4}x + \frac{15}{4}$
$f(x) = x + 4$

2) $f(x) = -4 + \frac{9}{4}x$
$g(x) = \frac{4}{9}x + \frac{16}{9}$

3) $h(n) = -2n^3 - 1$
$f(n) = \sqrt[3]{\frac{-n-1}{2}}$

4) $h(x) = \sqrt[3]{\frac{x+1}{2}}$
$f(x) = 2 + 2x^3$

5) $g(x) = \frac{1}{x+2} - 1$
$f(x) = \frac{1}{x-2} + 2$

6) $g(x) = -\frac{3}{-x+3}$
$f(x) = \frac{3}{x} + 3$

Find the inverse of each function.

7) $f(x) = \frac{-2+3x}{2}$

8) $f(x) = 5 + \frac{9}{5}x$

9) $f(x) = -(x-1)^3$

10) $f(x) = \sqrt[5]{x-3} - 2$

11) $g(n) = \frac{2}{n} - 3$

12) $f(x) = -\frac{3}{x-1}$

Find the inverse of each function. Then graph the function and its inverse.

13) $h(x) = 4 + \frac{7}{4}x$

14) $f(x) = \frac{2}{3}x + \frac{4}{3}$

15) $h(x) = \sqrt[3]{x} - 1$

16) $g(x) = -3 - 2x^3$

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Answers to HW #15 - Inverse Relations & Functions

1) No

2) Yes

3) Yes

4) No

5) No

6) Yes

7) $f^{-1}(x) = \frac{2x+2}{3}$

8) $f^{-1}(x) = \frac{5}{9}x - \frac{25}{9}$

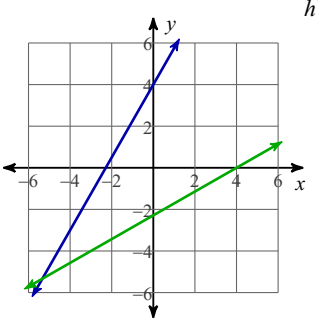
9) $f^{-1}(x) = -\sqrt[3]{x} + 1$

10) $f^{-1}(x) = 3 + (x+2)^5$

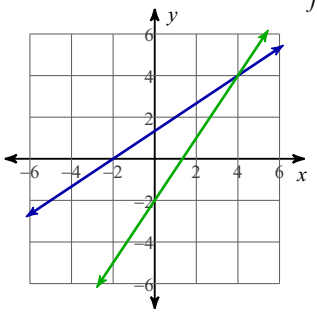
11) $g^{-1}(n) = \frac{2}{n+3}$

12) $f^{-1}(x) = -\frac{3}{x} + 1$

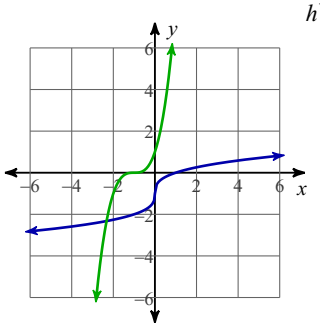
13) $h^{-1}(x) = \frac{4}{7}x - \frac{16}{7}$



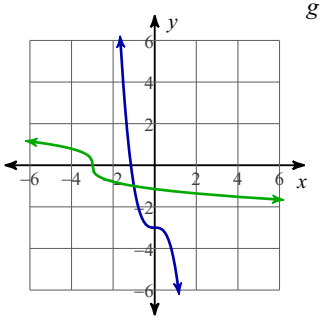
14) $f^{-1}(x) = -2 + \frac{3}{2}x$



15) $h^{-1}(x) = (x+1)^3$



16) $g^{-1}(x) = \sqrt[3]{\frac{-x-3}{2}}$



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